

Practice Exam #2

Full-Length Timed Simulation — Form B

110 questions • 5 hours 20 minutes

Instructions: NCEES FE Reference Handbook and approved calculator only. Pace 2.9 min/question. Write units at every step; never leave a blank.

Questions

1. A circular pipe of diameter $D = 0.25$ m carries a flow of $Q = 0.08$ m³/s. The average velocity is closest to:
(A) 1.0 m/s (B) 1.63 m/s
(C) 2.04 m/s (D) 0.41 m/s
2. A tank holds water to a depth of 6 m. The gauge pressure at the bottom is closest to:
(A) 6.0 kPa (B) 30 kPa
(C) 58.9 kPa (D) 117.7 kPa
3. Water ($\nu = 1.0 \times 10^{-6}$ m²/s) flows at $V = 2$ m/s in a $D = 0.10$ m pipe. The Reynolds number is:
(A) 2,000 (B) 20,000
(C) 100,000 (D) 200,000
4. Water discharges through a small orifice under a constant head of 5 m. Neglecting losses, the jet velocity is closest to:
(A) 9.9 m/s (B) 7.0 m/s
(C) 4.95 m/s (D) 14.0 m/s
5. For a pipe with $f = 0.02$, $L = 200$ m, $D = 0.30$ m, and $V = 3$ m/s, the Darcy–Weisbach head loss is closest to:
(A) 3.1 m (B) 6.1 m
(C) 12.2 m (D) 1.5 m
6. Using the Hazen–Williams equation (SI form $V = 0.849 C R_h^{0.63} S^{0.54}$) with $C = 120$, $R_h = 0.05$ m, and slope $S = 0.004$, the velocity is closest to:
(A) 0.78 m/s (B) 1.56 m/s
(C) 0.39 m/s (D) 1.20 m/s
7. An open channel has Manning $n = 0.013$, hydraulic radius $R_h = 0.5$ m, and slope $S = 0.001$. The velocity (SI Manning) is closest to:
(A) 0.77 m/s (B) 1.21 m/s
(C) 1.53 m/s (D) 3.06 m/s

8. A pump delivers $Q = 0.05 \text{ m}^3/\text{s}$ against a total head of 40 m at an efficiency of 75%. The required shaft power is closest to:
- (A) 19.6 kW (B) 14.7 kW
(C) 39.2 kW (D) 26.2 kW
9. A fitting has a minor-loss coefficient $K = 0.5$ where the velocity is 4 m/s. The head loss across the fitting is closest to:
- (A) 0.41 m (B) 0.82 m
(C) 0.20 m (D) 1.63 m
10. Two identical pumps, each delivering $0.03 \text{ m}^3/\text{s}$ at a given head, are placed in parallel at that head. The combined discharge is:
- (A) $0.03 \text{ m}^3/\text{s}$ (B) $0.06 \text{ m}^3/\text{s}$
(C) $0.09 \text{ m}^3/\text{s}$ (D) $0.015 \text{ m}^3/\text{s}$
11. A suppressed rectangular weir of length $L = 2 \text{ m}$ operates under a head $H = 0.3 \text{ m}$. Using $Q = 1.84 L H^{1.5}$, the discharge is closest to:
- (A) $1.10 \text{ m}^3/\text{s}$ (B) $0.30 \text{ m}^3/\text{s}$
(C) $0.60 \text{ m}^3/\text{s}$ (D) $2.21 \text{ m}^3/\text{s}$
12. A vertical rectangular gate 2 m wide extends from the water surface down to a depth of 3 m. The resultant hydrostatic force on the gate is closest to:
- (A) 88.3 kN (B) 176.6 kN
(C) 44.1 kN (D) 58.9 kN
13. In an open channel the flow has velocity $V = 2 \text{ m/s}$ at a depth $y = 0.5 \text{ m}$. The Froude number is closest to:
- (A) 0.45 (B) 0.90
(C) 1.28 (D) 1.80
14. A treatment unit reduces an influent concentration of 250 mg/L to 25 mg/L. The removal efficiency is:
- (A) 25% (B) 75%
(C) 89% (D) 90%
15. A tank of volume 2000 m^3 receives a flow of $500 \text{ m}^3/\text{d}$. The hydraulic detention time is closest to:
- (A) 4 h (B) 24 h
(C) 96 h (D) 48 h
16. A clarifier with surface area 200 m^2 treats $4000 \text{ m}^3/\text{d}$. The surface overflow rate is:
- (A) 20 m/d (B) 10 m/d
(C) 40 m/d (D) 800 m/d

17. A flow of $10,000 \text{ m}^3/\text{d}$ carries a BOD of 200 mg/L . The BOD mass loading is:
- (A) 200 kg/d (B) $2,000 \text{ kg/d}$
(C) $20,000 \text{ kg/d}$ (D) 20 kg/d
18. A water has calcium hardness of 80 mg/L as CaCO_3 and magnesium hardness of 30 mg/L as CaCO_3 . The total hardness is:
- (A) 50 mg/L (B) 80 mg/L
(C) 110 mg/L (D) 220 mg/L
19. A contact basin maintains a chlorine residual of 2 mg/L for 30 minutes. The CT value is:
- (A) 15 mg-min/L (B) 32 mg-min/L
(C) 28 mg-min/L (D) 60 mg-min/L
20. An activated-sludge reactor has $V = 2500 \text{ m}^3$, $\text{MLVSS} = 2500 \text{ mg/L}$, influent $Q = 5000 \text{ m}^3/\text{d}$ and $S_0 = 200 \text{ mg/L}$. The F/M ratio is closest to:
- (A) 0.16 d^{-1} (B) 0.32 d^{-1}
(C) 0.08 d^{-1} (D) 0.40 d^{-1}
21. An aeration basin holds $V = 4000 \text{ m}^3$ at $X = 3000 \text{ mg/L}$. Sludge is wasted at $Q_w = 50 \text{ m}^3/\text{d}$ with $X_w = 8000 \text{ mg/L}$ (neglect effluent solids). The solids retention time is closest to:
- (A) 10 d (B) 30 d
(C) 15 d (D) 60 d
22. A wastewater has ultimate BOD $L_0 = 300 \text{ mg/L}$ and a rate constant $k = 0.23 \text{ d}^{-1}$ (base e). The BOD exerted in 5 days is closest to:
- (A) 105 mg/L (B) 150 mg/L
(C) 205 mg/L (D) 95 mg/L
23. A water with chlorine demand 6 mg/L is dosed to leave a residual of 0.5 mg/L . For a flow of $3000 \text{ m}^3/\text{d}$, the chlorine feed rate is closest to:
- (A) 19.5 kg/d (B) 18.0 kg/d
(C) 1.95 kg/d (D) 6.5 kg/d
24. A process produces 1000 kg/d of dry solids in a sludge that is 3% solids (assume sludge specific gravity ≈ 1). The sludge volume is closest to:
- (A) $30 \text{ m}^3/\text{d}$ (B) $33.3 \text{ m}^3/\text{d}$
(C) $3.0 \text{ m}^3/\text{d}$ (D) $333 \text{ m}^3/\text{d}$
25. A water contains 100 mg/L of bicarbonate (HCO_3^- , equivalent weight 61). Expressed as CaCO_3 (equivalent weight 50), the alkalinity is closest to:
- (A) 100 mg/L (B) 50 mg/L
(C) 61 mg/L (D) 82 mg/L

26. A municipal wastewater carries a BOD load of 180 kg/d. At a typical per-capita contribution of 0.06 kg BOD/cap/d, the population equivalent is:
- (A) 1,000 (B) 1,800
(C) 3,000 (D) 300
27. Using the rational method (SI: $Q = CiA/360$, i in mm/hr, A in ha) with $C = 0.6$, $i = 50$ mm/hr, $A = 2$ ha, the peak flow is closest to:
- (A) $0.167 \text{ m}^3/\text{s}$ (B) $0.60 \text{ m}^3/\text{s}$
(C) $0.083 \text{ m}^3/\text{s}$ (D) $1.0 \text{ m}^3/\text{s}$
28. Using the rational method in U.S. units ($Q = CiA$, cfs) with $C = 0.5$, $i = 2$ in/hr, $A = 10$ ac, the peak flow is:
- (A) 5 cfs (B) 10 cfs
(C) 20 cfs (D) 100 cfs
29. Overland flow travels 300 m at an average velocity of 0.5 m/s. The travel (concentration) time is:
- (A) 5 min (B) 6 min
(C) 10 min (D) 600 min
30. A constant peak flow of $0.167 \text{ m}^3/\text{s}$ persists for 2 hours. The total runoff volume is closest to:
- (A) 600 m^3 (B) 400 m^3
(C) $1,200 \text{ m}^3$ (D) $1,202 \text{ m}^3$
31. A stream has a dissolved-oxygen saturation of 9 mg/L and an actual DO of 7 mg/L. The initial DO deficit is:
- (A) 2 mg/L (B) 9 mg/L
(C) 7 mg/L (D) 16 mg/L
32. Over a watershed, annual precipitation is 1000 mm, evapotranspiration 600 mm, and runoff 250 mm. The residual (infiltration plus storage change) is:
- (A) 100 mm (B) 150 mm
(C) 350 mm (D) 250 mm
33. For a Streeter–Phelps DO sag with $k_d = 0.3 \text{ d}^{-1}$, $k_r = 0.5 \text{ d}^{-1}$, initial deficit $D_0 = 2 \text{ mg/L}$, and initial ultimate BOD $L_a = 20 \text{ mg/L}$, the critical (time of minimum DO) is closest to:
- (A) 1.0 d (B) 1.8 d
(C) 2.18 d (D) 3.5 d
34. A rainfall depth of 25 mm falls over a 4-ha area. The rainfall volume is:
- (A) 100 m^3 (B) 250 m^3
(C) 500 m^3 (D) $1,000 \text{ m}^3$

35. Development raises the runoff coefficient from 0.2 to 0.7 over 5 ha for a design intensity of 40 mm/hr (SI rational, /360). The increase in peak runoff is closest to:
- (A) $0.278 \text{ m}^3/\text{s}$ (B) $0.556 \text{ m}^3/\text{s}$
(C) $0.139 \text{ m}^3/\text{s}$ (D) $1.0 \text{ m}^3/\text{s}$
36. A storm has a 1% probability of being equaled or exceeded in any year. Its return period is:
- (A) 10 yr (B) 100 yr
(C) 50 yr (D) 1,000 yr
37. An aquifer has $K = 1 \times 10^{-4} \text{ m/s}$, gradient 0.02, and cross-sectional area 10 m^2 . The Darcy flow rate is closest to:
- (A) $2 \times 10^{-4} \text{ m}^3/\text{s}$ (B) $1 \times 10^{-5} \text{ m}^3/\text{s}$
(C) $2 \times 10^{-5} \text{ m}^3/\text{s}$ (D) $1 \times 10^{-4} \text{ m}^3/\text{s}$
38. For the aquifer of the previous item, with effective porosity $n = 0.25$, the seepage (interstitial) velocity is closest to:
- (A) $8 \times 10^{-6} \text{ m/s}$ (B) $2 \times 10^{-6} \text{ m/s}$
(C) $5 \times 10^{-7} \text{ m/s}$ (D) $4 \times 10^{-6} \text{ m/s}$
39. Two wells 150 m apart show a head difference of 3 m. The hydraulic gradient is:
- (A) 0.5 (B) 0.02
(C) 0.05 (D) 0.2
40. A confined aquifer has hydraulic conductivity $K = 25 \text{ m/d}$ and thickness $b = 12 \text{ m}$. The transmissivity is:
- (A) $2.08 \text{ m}^2/\text{d}$ (B) $37 \text{ m}^2/\text{d}$
(C) $150 \text{ m}^2/\text{d}$ (D) $300 \text{ m}^2/\text{d}$
41. For steady confined radial flow (Thiem), $T = 300 \text{ m}^2/\text{d}$, head difference 5 m between $r_1 = 10 \text{ m}$ and $r_2 = 100 \text{ m}$. The well discharge is closest to:
- (A) $4,093 \text{ m}^3/\text{d}$ (B) $2,046 \text{ m}^3/\text{d}$
(C) $9,425 \text{ m}^3/\text{d}$ (D) $1,000 \text{ m}^3/\text{d}$
42. Groundwater flows at $Q = 0.001 \text{ m}^3/\text{s}$ through a cross-section of 5 m^2 . The specific discharge (Darcy velocity) is:
- (A) 0.005 m/s (B) 0.001 m/s
(C) 0.0002 m/s (D) 0.0005 m/s
43. A contaminant front moves with a seepage velocity of $8 \times 10^{-6} \text{ m/s}$. The time to travel 100 m is closest to:
- (A) 0.40 yr (B) 4.0 yr
(C) 40 yr (D) 0.04 yr

44. A soil sample has void volume 0.3 m^3 within a total volume of 1.0 m^3 . The porosity is:
- (A) 0.70 (B) 0.43
(C) 0.23 (D) 0.30
45. A sorbing solute has $K_d = 0.5 \text{ cm}^3/\text{g}$ in a soil with bulk density $\rho_b = 1.6 \text{ g/cm}^3$ and porosity $n = 0.4$. The retardation factor is:
- (A) 1.5 (B) 2.0
(C) 3.0 (D) 4.0
46. At 25°C and 1 atm, SO_2 (MW = 64) is present at 0.04 ppm by volume. Using molar volume 24.45 L/mol , the mass concentration is closest to:
- (A) $52 \text{ }\mu\text{g/m}^3$ (B) $105 \text{ }\mu\text{g/m}^3$
(C) $210 \text{ }\mu\text{g/m}^3$ (D) $26 \text{ }\mu\text{g/m}^3$
47. At 25°C and 1 atm, NO_2 (MW = 46) is measured at $100 \text{ }\mu\text{g/m}^3$. The volumetric mixing ratio is closest to:
- (A) 0.053 ppm (B) 0.106 ppm
(C) 0.027 ppm (D) 0.50 ppm
48. A process has an emission factor of 2 kg/ton and a throughput of 500 ton/day . The emission rate is:
- (A) 250 kg/d (B) 500 kg/d
(C) $1,000 \text{ kg/d}$ (D) $2,000 \text{ kg/d}$
49. Assuming the dry adiabatic lapse rate of 9.8°C/km , a parcel rising 2 km cools by:
- (A) 4.9°C (B) 9.8°C
(C) 14.7°C (D) 19.6°C
50. An electrostatic precipitator has collection area $A = 400 \text{ m}^2$, drift velocity $w = 0.1 \text{ m/s}$, and gas flow $Q = 50 \text{ m}^3/\text{s}$. By the Deutsch–Anderson equation, the efficiency is closest to:
- (A) 55.1% (B) 80%
(C) 45% (D) 99%
51. Which control device is best suited for collecting very fine (sub-micron) particulate matter at high efficiency?
- (A) Settling chamber (B) Single cyclone
(C) Fabric filter (baghouse) (D) Coarse screen
52. A cyclone separator removes particulate matter primarily by means of:
- (A) electrostatic charging (B) centrifugal (inertial) force
(C) chemical absorption (D) thermal condensation

53. A stack discharges $Q = 20 \text{ m}^3/\text{s}$ of gas containing a pollutant at 0.05 g/m^3 . The mass emission rate is:
- (A) 0.5 g/s (B) 1.0 g/s
(C) 2.0 g/s (D) 4.0 g/s
54. At 25°C and 1 atm (molar volume 24.45 L/mol), a gas sample occupying 48.9 L contains how many moles?
- (A) 1 mol (B) 1.5 mol
(C) 2.0 mol (D) 2.18 mol
55. A solution contains 100 mg/L of CaCO_3 ($\text{MW} = 100$). Its molar concentration is:
- (A) 0.001 M (B) 0.01 M
(C) 0.1 M (D) 0.0001 M
56. A solution has a hydrogen-ion concentration $[\text{H}^+] = 1 \times 10^{-4} \text{ mol/L}$. Its pH is:
- (A) 10 (B) 7
(C) 3 (D) 4
57. A solution has a pH of 9 at 25°C . Its pOH is:
- (A) 9 (B) 14
(C) 5 (D) 7
58. A first-order reaction has rate constant $k = 0.05 \text{ d}^{-1}$. Its half-life is closest to:
- (A) 6.9 d (B) 13.9 d
(C) 20 d (D) 3.5 d
59. For a compound with Henry's constant $K_H = 0.02 \text{ atm m}^3/\text{mol}$ and aqueous concentration 0.5 mol/m^3 , the equilibrium partial pressure is:
- (A) 0.01 atm (B) 0.02 atm
(C) 0.04 atm (D) 0.005 atm
60. For complete combustion of methane, $\text{CH}_4 + 2 \text{O}_2 \rightarrow \text{CO}_2 + 2 \text{H}_2\text{O}$, the moles of O_2 required to burn 4 mol of CH_4 is:
- (A) 2 (B) 4
(C) 6 (D) 8
61. A first-order decay starts at $C_0 = 100 \text{ mg/L}$ with $k = 0.1 \text{ hr}^{-1}$. After 10 hours the concentration is closest to:
- (A) 50 mg/L (B) 10 mg/L
(C) 36.8 mg/L (D) 90 mg/L

62. In dilute aqueous solution, a concentration of 15 mg/L is essentially equal to:
- (A) 1.5 ppm (B) 15 ppm
(C) 150 ppm (D) 1,500 ppm
63. A town of 20,000 people grows arithmetically at 500 persons/yr. Its projected population after 10 years is:
- (A) 20,500 (B) 22,500
(C) 25,000 (D) 30,000
64. A town of 20,000 people grows geometrically at 3%/yr. Its projected population after 10 years is closest to:
- (A) 26,878 (B) 26,000
(C) 25,000 (D) 27,000
65. A city of 30,000 people uses 400 L/capita/day. The average daily demand is:
- (A) 1,200 m³/d (B) 120 m³/d
(C) 1,200,000 m³/d (D) 12,000 m³/d
66. A first-order reaction ($k = 0.5 \text{ hr}^{-1}$) is to achieve 90% conversion in a single CSTR. The required hydraulic residence time is closest to:
- (A) 4.6 h (B) 18 h
(C) 9 h (D) 36 h
67. For the same reaction ($k = 0.5 \text{ hr}^{-1}$, 90% conversion) carried out in a plug-flow reactor, the required residence time is closest to:
- (A) 4.6 h (B) 18 h
(C) 9 h (D) 2.3 h
68. A tension member carries an axial load of 10,000 N over a cross-sectional area of 0.002 m². The normal stress is:
- (A) 0.5 MPa (B) 2 MPa
(C) 5 MPa (D) 50 MPa
69. A water system has an average demand of 12,000 m³/d and a peaking factor of 2.5. The peak-day demand is:
- (A) 4,800 m³/d (B) 12,000 m³/d
(C) 24,000 m³/d (D) 30,000 m³/d
70. A 1000-mm-long bar elongates 0.5 mm under load. The axial strain is:
- (A) 0.0005 (B) 0.005
(C) 0.05 (D) 0.0001

71. A community of 50,000 people generates 2 kg/capita/day of MSW. The daily generation is:
(A) 25 ton/d (B) 50 ton/d
(C) 100 ton/d (D) 1,000 ton/d
72. Refuse is compacted from a loose volume of 10 m^3 to 2.5 m^3 . The compaction ratio is:
(A) 2 (B) 4
(C) 8 (D) 0.25
73. A landfill receives 100 ton/d of refuse compacted to an in-place density of 600 kg/m^3 . The annual airspace consumed is closest to:
(A) $60,833 \text{ m}^3/\text{yr}$ (B) $6,083 \text{ m}^3/\text{yr}$
(C) $608,333 \text{ m}^3/\text{yr}$ (D) $36,500 \text{ m}^3/\text{yr}$
74. Landfill gas is generated at 1000 m^3 total with the typical $\approx 50/50$ methane-to-carbon-dioxide split. The methane volume is:
(A) 250 m^3 (B) $1,000 \text{ m}^3$
(C) 750 m^3 (D) 500 m^3
75. Compacting refuse from 10 m^3 to 2.5 m^3 corresponds to a volume reduction of:
(A) 25% (B) 40%
(C) 75% (D) 90%
76. Under federal law, the cradle-to-grave management of *actively generated* hazardous waste is governed primarily by:
(A) RCRA (B) CERCLA (Superfund)
(C) the Clean Air Act (D) NEPA
77. A waste exhibiting the characteristics of ignitability, corrosivity, reactivity, or toxicity is classified as hazardous. These are the four “characteristic” criteria abbreviated:
(A) BTEX (B) TCLP
(C) ICRT (D) PCB
78. Of 200 ton of MSW generated, 60 ton is recycled. The diversion (recycling) rate is:
(A) 30% (B) 40%
(C) 60% (D) 3%
79. The larger root of $x^2 - 5x + 6 = 0$ is:
(A) 2 (B) 3
(C) 6 (D) 5
80. The derivative of $f(x) = 3x^2$ evaluated at $x = 4$ is:
(A) 12 (B) 16
(C) 48 (D) 24

81. The value of $\int_0^3 2x \, dx$ is:
- (A) 3 (B) 6
(C) 9 (D) 18
82. The value of $\sin 30^\circ$ is:
- (A) 0.5 (B) 0.866
(C) 1.0 (D) 0.707
83. The distance between the points $(0, 0)$ and $(3, 4)$ is:
- (A) 3.5 (B) 5
(C) 7 (D) 25
84. The value of x satisfying $10^x = 1000$ is:
- (A) 1 (B) 10
(C) 100 (D) 3
85. Under the NCEES Model Rules, the engineer's foremost obligation is to:
- (A) the client's budget (B) the employer's profit
(C) the safety, health, and welfare of the public (D) the engineer's reputation
86. An engineer should accept assignments only in areas of their:
- (A) highest pay (B) personal interest
(C) competence (D) seniority
87. The federal statute that establishes "waters of the United States" discharge permitting (NPDES) is the:
- (A) Clean Water Act (B) Clean Air Act
(C) Safe Drinking Water Act (D) RCRA
88. National primary drinking-water standards (e.g., MCLs at the tap) are set under the:
- (A) Clean Water Act (B) Clean Air Act
(C) Safe Drinking Water Act (D) CERCLA
89. The statute requiring an Environmental Impact Statement for major federal actions significantly affecting the environment is:
- (A) NEPA (B) OSHA Act
(C) RCRA (D) FIFRA

90. When an engineer discovers that a decision by a supervisor endangers the public, the ethically required action is to:
- (A) follow orders without comment (B) quietly resign
(C) notify the proper authority and document the concern (D) ignore it if not personally liable
91. The present worth of \$10,000 received 5 years from now at $i = 6\%$ is closest to:
- (A) \$9,434 (B) \$5,000
(C) \$7,473 (D) \$13,382
92. The present worth of a 10-year uniform series of \$2,000/yr at $i = 8\%$ is closest to:
- (A) \$13,420 (B) \$20,000
(C) \$10,000 (D) \$15,000
93. A \$50,000 loan is repaid by equal annual payments over 20 years at $i = 10\%$. Each payment is closest to:
- (A) \$2,500 (B) \$5,873
(C) \$5,000 (D) \$10,000
94. A project has present-worth benefits of \$120,000 and costs of \$80,000. The benefit–cost ratio is:
- (A) 0.67 (B) 1.0
(C) 1.2 (D) 1.5
95. An asset costs \$100,000 with a salvage value of \$20,000 and a 8-year life. The straight-line annual depreciation is:
- (A) \$12,500 (B) \$2,500
(C) \$10,000 (D) \$80,000
96. A product sells for \$25/unit with variable cost \$15/unit and fixed cost \$50,000. The break-even quantity is:
- (A) 5,000 units (B) 2,000 units
(C) 3,333 units (D) 10,000 units
97. The mean of the data set $\{4, 8, 10, 6, 12\}$ is:
- (A) 6 (B) 8
(C) 10 (D) 40
98. The sample standard deviation of $\{2, 4, 6, 8, 10\}$ is closest to:
- (A) 2.83 (B) 2.0
(C) 3.16 (D) 4.0

99. A value of 85 comes from a distribution with mean 70 and standard deviation 10. Its z -score is:
- (A) 1.5 (B) 0.85
(C) 15 (D) 0.15
100. For a normal distribution, the approximate fraction of values within one standard deviation of the mean is:
- (A) 95% (B) 99.7%
(C) 50% (D) 68%
101. A 70-kg adult drinks 2 L/day of water containing a contaminant at 0.05 mg/L (chronic exposure, averaging factors ≈ 1). The chronic daily intake is closest to:
- (A) 0.0007 mg/kg-d (B) 0.00143 mg/kg-d
(C) 0.0014 mg/kg-d (D) 0.005 mg/kg-d
102. A chemical with chronic daily intake 0.0014 mg/kg-d has a cancer slope factor of $0.5 \text{ (mg/kg-d)}^{-1}$. The incremental lifetime cancer risk is closest to:
- (A) 7×10^{-4} (B) 1.4×10^{-3}
(C) 0.5 (D) 3.5×10^{-4}
103. A noncarcinogen has an intake of 0.02 mg/kg-d and a reference dose of 0.04 mg/kg-d. The hazard quotient is:
- (A) 2 (B) 0.8
(C) 0.5 (D) 0.04
104. An incremental lifetime cancer risk of 2×10^{-5} applies to a population of 50,000. The expected number of excess cancers is:
- (A) 5 (B) 10
(C) 0.5 (D) 1
105. A household uses 1000 kWh of electricity at a grid emission factor of 0.5 kg CO₂/kWh. The carbon footprint is:
- (A) 200 kg (B) 1,000 kg
(C) 500 kg (D) 50 kg
106. Using a global-warming potential of 25 for methane, the CO₂-equivalent of 10 ton of CH₄ is:
- (A) 250 ton (B) 25 ton
(C) 100 ton (D) 10 ton
107. Burning 100 L of gasoline emits about 2.3 kg CO₂ per liter. The total CO₂ is closest to:
- (A) 23 kg (B) 230 kg
(C) 2,300 kg (D) 100 kg

108. For an ideal gas with $n = 2$ mol at $T = 300$ K and $P = 100$ kPa ($R = 8.314$ J/mol K), the volume is closest to:
- (A) 0.025 m^3 (B) 0.10 m^3
 (C) 0.075 m^3 (D) 0.0499 m^3
109. A Carnot engine operates between a hot reservoir at 600 K and a cold reservoir at 300 K. Its maximum thermal efficiency is:
- (A) 0.50 (B) 0.25
 (C) 0.75 (D) 0.33
110. A closed system receives 100 kJ of heat and does 30 kJ of work on its surroundings. By the first law, the change in internal energy is:
- (A) 130 kJ (B) 30 kJ
 (C) 70 kJ (D) 100 kJ

Answer Key

1	2	3	4	5	6	7	8	9	10
B	C	D	A	B	A	C	D	A	B

11	12	13	14	15	16	17	18	19	20
C	A	B	D	C	A	B	C	D	A

21	22	23	24	25	26	27	28	29	30
B	C	A	B	D	C	A	B	C	D

31	32	33	34	35	36	37	38	39	40
A	B	C	D	A	B	C	A	B	D

41	42	43	44	45	46	47	48	49	50
A	C	A	D	C	B	A	C	D	A

51	52	53	54	55	56	57	58	59	60
C	C	B	C	A	D	C	B	A	D

61	62	63	64	65	66	67	68	69	70
C	B	C	A	D	B	A	C	D	A

71	72	73	74	75	76	77	78	79	80
C	B	A	D	C	A	C	A	B	D

81	82	83	84	85	86	87	88	89	90
C	A	B	D	C	C	A	C	A	C

91	92	93	94	95	96	97	98	99	100
C	A	B	D	C	A	B	C	A	D

101	102	103	104	105	106	107	108	109	110
B	A	C	D	C	A	B	D	A	C

Worked Solutions

1. **Answer: B.** $A = \frac{\pi}{4}(0.25)^2 = 0.0491 \text{ m}^2$; $V = Q/A = 0.08/0.0491 = 1.63 \text{ m/s}$. Handbook: Fluid Mechanics — continuity $Q = VA$

2. **Answer: C.** $p = \rho gh = 1000(9.81)(6) = 58,860 \text{ Pa} = 58.9 \text{ kPa}$. Handbook: Fluid Mechanics — hydrostatic pressure
3. **Answer: D.** $Re = VD/\nu = (2)(0.10)/(1.0 \times 10^{-6}) = 2 \times 10^5$. Handbook: Fluid Mechanics — Reynolds number
4. **Answer: A.** Torricelli: $V = \sqrt{2gh} = \sqrt{2(9.81)(5)} = 9.9 \text{ m/s}$. Handbook: Fluid Mechanics — orifice / Bernoulli
5. **Answer: B.** $h_f = f \frac{L}{D} \frac{V^2}{2g} = 0.02 \frac{200}{0.30} \frac{3^2}{2(9.81)} = 6.1 \text{ m}$. Handbook: Fluid Mechanics — Darcy–Weisbach
6. **Answer: A.** $V = 0.849(120)(0.05)^{0.63}(0.004)^{0.54} = 0.78 \text{ m/s}$. Handbook: Fluid Mechanics — Hazen–Williams
7. **Answer: C.** $V = \frac{1}{n} R_h^{2/3} S^{1/2} = \frac{1}{0.013} (0.5)^{2/3} (0.001)^{1/2} = 1.53 \text{ m/s}$. Handbook: Fluid Mechanics — Manning equation
8. **Answer: D.** $P = \frac{\rho g Q H}{\eta} = \frac{1000(9.81)(0.05)(40)}{0.75} = 26,160 \text{ W} = 26.2 \text{ kW}$. Handbook: Fluid Mechanics — pump power
9. **Answer: A.** $h_L = K \frac{V^2}{2g} = 0.5 \frac{4^2}{2(9.81)} = 0.41 \text{ m}$. Handbook: Fluid Mechanics — minor losses
10. **Answer: B.** Pumps in parallel add flow at common head: $Q = 2(0.03) = 0.06 \text{ m}^3/\text{s}$. Handbook: Fluid Mechanics — pumps in parallel
11. **Answer: C.** $Q = 1.84 L H^{1.5} = 1.84(2)(0.3)^{1.5} = 0.60 \text{ m}^3/\text{s}$. Handbook: Fluid Mechanics — rectangular weir
12. **Answer: A.** $h_c = 1.5 \text{ m}$, $A = 2(3) = 6 \text{ m}^2$; $F = \rho g h_c A = 1000(9.81)(1.5)(6) = 88,290 \text{ N} = 88.3 \text{ kN}$. Handbook: Fluid Mechanics — hydrostatic force on plane
13. **Answer: B.** $Fr = \frac{V}{\sqrt{gy}} = \frac{2}{\sqrt{9.81(0.5)}} = 0.90$. Handbook: Fluid Mechanics — Froude number
14. **Answer: D.** $E = \frac{250 - 25}{250} \times 100 = 90\%$. Handbook: Water/Wastewater — removal efficiency
15. **Answer: C.** $t = V/Q = 2000/500 = 4 \text{ d} \times 24 = 96 \text{ h}$. Handbook: Water/Wastewater — detention time
16. **Answer: A.** Overflow rate $= Q/A = 4000/200 = 20 \text{ m/d}$. Handbook: Water/Wastewater — surface overflow rate
17. **Answer: B.** Load $= Q C = 10,000 \text{ m}^3/\text{d} \times 200 \text{ g/m}^3 \div 1000 = 2,000 \text{ kg/d}$. Handbook: Water/Wastewater — mass loading
18. **Answer: C.** Total hardness $= 80 + 30 = 110 \text{ mg/L as CaCO}_3$. Handbook: Water/Wastewater — hardness

19. **Answer: D.** $CT = C \times t = 2 \text{ mg/L} \times 30 \text{ min} = 60 \text{ mg-min/L}$. Handbook: Water/Wastewater — disinfection CT
20. **Answer: A.** $F/M = \frac{QS_0}{VX} = \frac{5000(200)}{2500(2500)} = 0.16 \text{ d}^{-1}$. Handbook: Water/Wastewater — F/M ratio
21. **Answer: B.** $\theta_c = \frac{VX}{Q_w X_w} = \frac{4000(3000)}{50(8000)} = 30 \text{ d}$. Handbook: Water/Wastewater — solids retention time
22. **Answer: C.** $\text{BOD}_5 = L_0(1 - e^{-kt}) = 300(1 - e^{-0.23(5)}) = 300(0.683) = 205 \text{ mg/L}$. Handbook: Water/Wastewater — BOD decay
23. **Answer: A.** $\text{Dose} = 6 + 0.5 = 6.5 \text{ mg/L}$; $\text{feed} = 6.5(3000)/1000 = 19.5 \text{ kg/d}$. Handbook: Water/Wastewater — chlorine dose
24. **Answer: B.** $V = \frac{\text{dry solids}}{(\% \text{ solids})\rho} = \frac{1000}{0.03(1000)} = 33.3 \text{ m}^3/\text{d}$. Handbook: Water/Wastewater — sludge volume
25. **Answer: D.** As CaCO_3 : $100 \times \frac{50}{61} = 82 \text{ mg/L}$. Handbook: Water/Wastewater — alkalinity conversion
26. **Answer: C.** $\text{PE} = \frac{180 \text{ kg/d}}{0.06 \text{ kg/cap-d}} = 3,000 \text{ people}$. Handbook: Water/Wastewater — population equivalent
27. **Answer: A.** $Q = \frac{CiA}{360} = \frac{0.6(50)(2)}{360} = 0.167 \text{ m}^3/\text{s}$. Handbook: Hydrology — rational method (SI)
28. **Answer: B.** $Q = CiA = 0.5(2)(10) = 10 \text{ cfs}$. Handbook: Hydrology — rational method (U.S.)
29. **Answer: C.** $t = L/v = 300/0.5 = 600 \text{ s} = 10 \text{ min}$. Handbook: Hydrology — time of concentration
30. **Answer: D.** $V = Qt = 0.167(7200 \text{ s}) = 1,202 \text{ m}^3$. Handbook: Hydrology — runoff volume
31. **Answer: A.** $D_0 = \text{DO}_{\text{sat}} - \text{DO} = 9 - 7 = 2 \text{ mg/L}$. Handbook: Hydrology — DO deficit
32. **Answer: B.** $\text{Residual} = P - ET - R = 1000 - 600 - 250 = 150 \text{ mm}$. Handbook: Hydrology — water budget
33. **Answer: C.** $t_c = \frac{1}{k_r - k_d} \ln \left[\frac{k_r}{k_d} \left(1 - \frac{D_0(k_r - k_d)}{k_d L_a} \right) \right] = \frac{1}{0.2} \ln \left[\frac{0.5}{0.3} \left(1 - \frac{2(0.2)}{0.3(20)} \right) \right] = 2.18 \text{ d}$. Handbook: Hydrology — Streeter-Phelps critical time
34. **Answer: D.** $V = (0.025 \text{ m})(4 \times 10,000 \text{ m}^2) = 1,000 \text{ m}^3$. Handbook: Hydrology — depth-to-volume
35. **Answer: A.** $\Delta Q = \frac{(0.7 - 0.2)(40)(5)}{360} = 0.278 \text{ m}^3/\text{s}$. Handbook: Hydrology — rational method change

36. **Answer: B.** Return period $T = 1/p = 1/0.01 = 100$ yr. Handbook: Hydrology — return period
37. **Answer: C.** $Q = KiA = (1 \times 10^{-4})(0.02)(10) = 2 \times 10^{-5} \text{ m}^3/\text{s}$. Handbook: Groundwater — Darcy's law
38. **Answer: A.** $v_d = Ki = 2 \times 10^{-6} \text{ m/s}$; $v_s = v_d/n = 2 \times 10^{-6}/0.25 = 8 \times 10^{-6} \text{ m/s}$. Handbook: Groundwater — seepage velocity
39. **Answer: B.** $i = \Delta h/L = 3/150 = 0.02$. Handbook: Groundwater — hydraulic gradient
40. **Answer: D.** $T = Kb = 25(12) = 300 \text{ m}^2/\text{d}$. Handbook: Groundwater — transmissivity
41. **Answer: A.** $Q = \frac{2\pi T(h_2 - h_1)}{\ln(r_2/r_1)} = \frac{2\pi(300)(5)}{\ln(100/10)} = 4,093 \text{ m}^3/\text{d}$. Handbook: Groundwater — Thiem equation
42. **Answer: C.** $q = Q/A = 0.001/5 = 0.0002 \text{ m/s}$. Handbook: Groundwater — specific discharge
43. **Answer: A.** $t = L/v = 100/(8 \times 10^{-6}) = 1.25 \times 10^7 \text{ s} = 0.40 \text{ yr}$. Handbook: Groundwater — travel time
44. **Answer: D.** $n = V_v/V_t = 0.3/1.0 = 0.30$. Handbook: Soils — porosity
45. **Answer: C.** $R = 1 + \frac{\rho_b}{n}K_d = 1 + \frac{1.6}{0.4}(0.5) = 3.0$. Handbook: Soils — retardation factor
46. **Answer: B.** $C = \frac{\text{ppm} \cdot MW \cdot 1000}{24.45} = \frac{0.04(64)(1000)}{24.45} = 105 \text{ } \mu\text{g}/\text{m}^3$. Handbook: Air Quality — ppm to $\mu\text{g}/\text{m}^3$
47. **Answer: A.** $\text{ppm} = \frac{C \cdot 24.45}{MW \cdot 1000} = \frac{100(24.45)}{46(1000)} = 0.053 \text{ ppm}$. Handbook: Air Quality — $\mu\text{g}/\text{m}^3$ to ppm
48. **Answer: C.** Rate = $(2 \text{ kg/ton})(500 \text{ ton/d}) = 1,000 \text{ kg/d}$. Handbook: Air Quality — emission factor
49. **Answer: D.** $\Delta T = (9.8 \text{ }^\circ\text{C/km})(2 \text{ km}) = 19.6 \text{ }^\circ\text{C}$. Handbook: Air Quality — adiabatic lapse rate
50. **Answer: A.** $\eta = 1 - e^{-wA/Q} = 1 - e^{-0.1(400)/50} = 1 - e^{-0.8} = 0.551 = 55.1\%$. Handbook: Air Quality — Deutsch-Anderson ESP
51. **Answer: C.** A fabric filter (baghouse) collects fine, sub-micron particulate at high efficiency; settling chambers and single cyclones capture only coarse particles. Handbook: Air Quality — particulate control devices
52. **Answer: C.** A cyclone uses centrifugal (inertial) force to fling particles to the wall; it is effective for coarse particulate but not sub-micron PM. Handbook: Air Quality — cyclone separator
53. **Answer: B.** $E = QC = 20 \text{ m}^3/\text{s} \times 0.05 \text{ g}/\text{m}^3 = 1.0 \text{ g/s}$. Handbook: Air Quality — mass emission rate

54. **Answer: C.** $n = V/24.45 = 48.9/24.45 = 2.0$ mol. Handbook: Air Quality — ideal-gas molar volume
55. **Answer: A.** $M = \frac{100 \text{ mg/L}}{100 \text{ g/mol} \times 1000 \text{ mg/g}} = 0.001$ M. Handbook: Chemistry — mg/L to molarity
56. **Answer: D.** $\text{pH} = -\log_{10}[\text{H}^+] = -\log_{10}(10^{-4}) = 4$. Handbook: Chemistry — pH
57. **Answer: C.** $\text{pOH} = 14 - \text{pH} = 14 - 9 = 5$. Handbook: Chemistry — pH/pOH
58. **Answer: B.** $t_{1/2} = 0.693/k = 0.693/0.05 = 13.9$ d. Handbook: Chemistry — first-order half-life
59. **Answer: A.** $p = K_H C = 0.02 \text{ atm m}^3/\text{mol} \times 0.5 \text{ mol/m}^3 = 0.01$ atm. Handbook: Chemistry — Henry's law
60. **Answer: D.** Per mole CH_4 needs 2 mol O_2 ; for 4 mol: $2(4) = 8$ mol. Handbook: Chemistry — combustion stoichiometry
61. **Answer: C.** $C = C_0 e^{-kt} = 100 e^{-0.1(10)} = 100(0.368) = 36.8$ mg/L. Handbook: Chemistry — first-order kinetics
62. **Answer: B.** In dilute water $1 \text{ mg/L} \approx 1 \text{ ppm}$, so $15 \text{ mg/L} = 15 \text{ ppm}$. Handbook: Chemistry — mg/L vs ppm
63. **Answer: C.** $P = P_0 + k_a t = 20,000 + 500(10) = 25,000$. Handbook: Fundamentals — arithmetic projection
64. **Answer: A.** $P = P_0(1+r)^n = 20,000(1.03)^{10} = 26,878$. Handbook: Fundamentals — geometric projection
65. **Answer: D.** $Q = 30,000 \times 400 \text{ L/d} = 1.2 \times 10^7 \text{ L/d} = 12,000 \text{ m}^3/\text{d}$. Handbook: Fundamentals — per-capita demand
66. **Answer: B.** CSTR first order: $\frac{C}{C_0} = \frac{1}{1+k\tau}$; for 0.1, $k\tau = 9$, $\tau = 9/0.5 = 18$ h. Handbook: Fundamentals — CSTR
67. **Answer: A.** PFR first order: $\tau = \frac{\ln(C_0/C)}{k} = \frac{\ln 10}{0.5} = 4.6$ h. Handbook: Fundamentals — PFR
68. **Answer: C.** $\sigma = F/A = 10,000/0.002 = 5 \times 10^6 \text{ Pa} = 5 \text{ MPa}$. Handbook: Fundamentals — normal stress
69. **Answer: D.** Peak $= 2.5 \times 12,000 = 30,000 \text{ m}^3/\text{d}$. Handbook: Fundamentals — peaking factor
70. **Answer: A.** $\varepsilon = \Delta L/L = 0.5/1000 = 5 \times 10^{-4}$. Handbook: Fundamentals — axial strain
71. **Answer: C.** $50,000 \times 2 \text{ kg/d} = 100,000 \text{ kg/d} = 100 \text{ ton/d}$. Handbook: Solid Waste — MSW generation
72. **Answer: B.** Compaction ratio $= V_{\text{loose}}/V_{\text{comp}} = 10/2.5 = 4$. Handbook: Solid Waste — compaction ratio

73. **Answer: A.** $\text{Mass/yr} = 100(1000)(365) = 3.65 \times 10^7 \text{ kg}$; $V = 3.65 \times 10^7 / 600 = 60,833 \text{ m}^3/\text{yr}$.
Handbook: Solid Waste — landfill airspace
74. **Answer: D.** $\approx 50\%$ of 1000 m^3 is $\text{CH}_4 = 500 \text{ m}^3$. Handbook: Solid Waste — landfill gas
75. **Answer: C.** $\text{Reduction} = \frac{10 - 2.5}{10} \times 100 = 75\%$. Handbook: Solid Waste — volume reduction
76. **Answer: A.** RCRA governs cradle-to-grave management of actively generated hazardous waste; CERCLA addresses abandoned/historic sites. Handbook: Hazardous Waste — RCRA vs CERCLA
77. **Answer: C.** Ignitability, Corrosivity, Reactivity, Toxicity $\rightarrow$ ICRT. Handbook: Hazardous Waste — characteristic wastes
78. **Answer: A.** $\text{Diversion} = 60/200 \times 100 = 30\%$. Handbook: Solid Waste — recycling diversion
79. **Answer: B.** $x^2 - 5x + 6 = (x - 2)(x - 3) = 0 \Rightarrow x = 2, 3$; larger is 3. Handbook: Mathematics — quadratic roots
80. **Answer: D.** $f'(x) = 6x$; at $x = 4$, $f' = 24$. Handbook: Mathematics — derivative
81. **Answer: C.** $\int_0^3 2x \, dx = x^2 \Big|_0^3 = 9$. Handbook: Mathematics — definite integral
82. **Answer: A.** $\sin 30^\circ = 0.5$. Handbook: Mathematics — trigonometry
83. **Answer: B.** $d = \sqrt{3^2 + 4^2} = 5$. Handbook: Mathematics — analytic geometry
84. **Answer: D.** $10^x = 1000 = 10^3 \Rightarrow x = 3$. Handbook: Mathematics — logarithms
85. **Answer: C.** The Model Rules hold the safety, health, and welfare of the public paramount. Handbook: Ethics — public welfare paramount
86. **Answer: C.** Engineers must perform services only in their areas of competence. Handbook: Ethics — competence
87. **Answer: A.** The Clean Water Act establishes NPDES discharge permitting. Handbook: Ethics/Regulation — CWA/NPDES
88. **Answer: C.** The Safe Drinking Water Act sets national primary drinking-water standards (MCLs). Handbook: Ethics/Regulation — SDWA
89. **Answer: A.** NEPA requires an Environmental Impact Statement for major federal actions. Handbook: Ethics/Regulation — NEPA
90. **Answer: C.** The engineer must notify the proper authority and document the safety concern. Handbook: Ethics — duty to report
91. **Answer: C.** $P = F(1 + i)^{-n} = 10,000(1.06)^{-5} = \$7,473$. Handbook: Economics — single-payment P/F
92. **Answer: A.** $P = A \frac{(1 + i)^n - 1}{i(1 + i)^n} = 2,000(6.710) = \$13,420$. Handbook: Economics — uniform-series P/A

93. **Answer: B.** $A = P \frac{i(1+i)^n}{(1+i)^n - 1} = 50,000(0.11746) = \$5,873$. Handbook: Economics — capital recovery A/P
94. **Answer: D.** $B/C = 120,000/80,000 = 1.5$. Handbook: Economics — benefit-cost ratio
95. **Answer: C.** $D = \frac{C - S}{n} = \frac{100,000 - 20,000}{8} = \$10,000/\text{yr}$. Handbook: Economics — straight-line depreciation
96. **Answer: A.** $Q^* = \frac{F}{p - v} = \frac{50,000}{25 - 15} = 5,000$ units. Handbook: Economics — break-even
97. **Answer: B.** $\bar{x} = \frac{4 + 8 + 10 + 6 + 12}{5} = \frac{40}{5} = 8$. Handbook: Statistics — mean
98. **Answer: C.** Mean = 6; $s = \sqrt{\frac{16 + 4 + 0 + 4 + 16}{5 - 1}} = \sqrt{10} = 3.16$. Handbook: Statistics — sample standard deviation
99. **Answer: A.** $z = \frac{x - \mu}{\sigma} = \frac{85 - 70}{10} = 1.5$. Handbook: Statistics — z-score
100. **Answer: D.** The empirical rule gives $\approx 68\%$ within one standard deviation. Handbook: Statistics — normal distribution
101. **Answer: B.** $\text{CDI} = \frac{C \cdot IR}{BW} = \frac{0.05(2)}{70} = 0.00143$ mg/kg-d. Handbook: Risk — chronic daily intake
102. **Answer: A.** Risk = $\text{CDI} \times SF = 0.0014(0.5) = 7 \times 10^{-4}$. Handbook: Risk — cancer risk
103. **Answer: C.** $HQ = \frac{\text{intake}}{RfD} = \frac{0.02}{0.04} = 0.5$. Handbook: Risk — hazard quotient
104. **Answer: D.** Cases = $(2 \times 10^{-5})(50,000) = 1$. Handbook: Risk — population risk
105. **Answer: C.** $1000 \text{ kWh} \times 0.5 \text{ kg/kWh} = 500 \text{ kg CO}_2$. Handbook: Energy — carbon footprint
106. **Answer: A.** $\text{CO}_2\text{e} = 10 \text{ ton} \times 25 = 250 \text{ ton}$. Handbook: Energy — GWP / CO_2 equivalent
107. **Answer: B.** $100 \text{ L} \times 2.3 \text{ kg/L} = 230 \text{ kg CO}_2$. Handbook: Energy — fuel combustion CO_2
108. **Answer: D.** $V = \frac{nRT}{P} = \frac{2(8.314)(300)}{100,000} = 0.0499 \text{ m}^3$. Handbook: Thermodynamics — ideal gas law
109. **Answer: A.** $\eta = 1 - \frac{T_c}{T_h} = 1 - \frac{300}{600} = 0.50$. Handbook: Thermodynamics — Carnot efficiency
110. **Answer: C.** First law: $\Delta U = Q - W = 100 - 30 = 70 \text{ kJ}$. Handbook: Thermodynamics — first law